

Recall from last class:

- If V is spanned by a finite set, then V is called **finite dimensional**, and the **dimension** of V , written $\dim(V)$, is the number of vectors in a basis of V . Define $\dim(\{0_V\}) = 0$. If V is not spanned by any finite set, then V is called **infinite dimensional**.
- **Theorem:** Let H be a subspace of a finite dimensional vector space V . Any linearly independent set in H can be expanded to a basis for H , and H is finite dimensional and $\dim(H) \leq \dim(V)$.
- **Basis Theorem:** Let V be an n -dimensional vector space with $n \geq 1$. Any linearly independent set of exactly n vectors in V is a basis for V . Any set of exactly n vectors that spans V is a basis for V .
- Let A be an $m \times n$ matrix with rows $a_1, \dots, a_m$. Define the **row space** of A by

$$\text{Row}(A) = \text{Span}\{a_1, \dots, a_m\}.$$

- **Theorem:** If A is row equivalent to B , then $\text{Row}(A) = \text{Row}(B)$. Furthermore, if B is in reduced row echelon form, the non-zero rows of B form a basis for $\text{Row}(A)$ and $\text{Row}(B)$.
- The **rank** of the $m \times n$ matrix A is the dimension of $\text{Col}(A)$. We write $\text{rank}(A) = \dim(\text{Col}(A))$. The **nullity** of A is the dimension of $N(A)$.
- **Rank-Nullity Theorem:** Let A be an $m \times n$ matrix.

1.

$$\dim \text{Row}(A) = \dim \text{Col}(A)$$

2.

$$\text{rank}(A) + \dim N(A) = n$$

- **Invertible Matrix Theorem Continued** The following are equivalent:
 - m. The columns of A form a basis of $\mathbb{R}^n$.
 - n. $\text{Col}(A) = \mathbb{R}^n$.
 - o. $\dim \text{Col}(A) = n$.
 - p. $\text{rank}(A) = n$.
 - q. $N(A) = \{0\}$.
 - r. $\dim N(A) = 0$.
- **Theorem:** Let $T : V \rightarrow W$ be a linear transformation between vector spaces. If V is finite dimensional, then

$$\dim \text{im}(T) + \dim \ker(T) = \dim(V).$$

Rank-Nullity Theorem: Let A be an $m \times n$ matrix.

a.

$$\dim \text{Row}(A) = \dim \text{Col}(A)$$

b.

$$\text{rank}(A) + \dim N(A) = n$$

1.

(a) Let A be a 7 by 5 matrix. What is the largest possible rank?

(b) Let A be a 3 by 7 matrix. What is the smallest possible nullity?

(c) Let A be a 5 by 6 matrix. Suppose the solutions to $Ax = 0$ are all multiple of one non-zero vector. Will the matrix equation $Ax = b$ always be consistent?

Invertible Matrix Theorem Continued The following are equivalent to A being invertible:

m. The columns of A form a basis of $\mathbb{R}^n$.

n. $\text{Col}(A) = \mathbb{R}^n$.

o. $\dim \text{Col}(A) = n$.

p. $\text{rank}(A) = n$.

q. $N(A) = \{0\}$.

r. $\dim N(A) = 0$.

Theorem: Let $T : V \rightarrow W$ be a linear transformation between vector spaces. If V is finite dimensional, then

$$\dim \text{im}(T) + \dim \ker(T) = \dim(V).$$

2. Let $\mathcal{B} = \{b_1, b_2\}$ and $\mathcal{C} = \{c_1, c_2\}$ be bases for a vector space V where $b_1 = c_1 + c_2$ and $b_2 = -c_1 + c_2$. How can we change from the basis $\mathcal{B}$ to $\mathcal{C}$?

Theorem: Let $\mathcal{B} = \{b_1, \dots, b_n\}$ and $\mathcal{C} = \{c_1, \dots, c_n\}$ be bases of a vector space V . Then there exists a unique $n \times n$ matrix P such that $[x]_{\mathcal{C}} = P[x]_{\mathcal{B}}$.

3. Prove the theorem.

Note: We call P the **change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$** . In general

$$P = [[b_1]_{\mathcal{C}} \ \dots \ [b_n]_{\mathcal{C}}].$$

Also P^{-1} is the change of coordinates matrix from $\mathcal{C}$ to $\mathcal{B}$.

4. Let $b_1 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$, $b_2 = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $c_1 = \begin{bmatrix} -7 \\ 9 \end{bmatrix}$, and $c_2 = \begin{bmatrix} -5 \\ 7 \end{bmatrix}$. Find the change of coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$.

5. Let $\mathcal{B} = \{1 - 2t + t^2, 3 - 5t + 4t^2, 2t + 3t^2\}$ and $\mathcal{E} = \{1, t, t^2\}$ be bases for $\mathbb{P}_2$. Find the change of coordinates matrix from $\mathcal{B}$ to $\mathcal{E}$.

6. Find $[-1 + 2t]_{\mathcal{B}}$.

Let V be a vector space with basis $\mathcal{B} = \{b_1, \dots, b_n\}$ and W be a vector space with basis $\mathcal{C} = \{c_1, \dots, c_m\}$. Given a linear transformation $T : V \rightarrow W$, we naturally seek an associated matrix to represent T using the bases; namely we want an $m \times n$ matrix M such that $[T(x)]_{\mathcal{C}} = M[x]_{\mathcal{B}}$ for all $x \in V$. We call M the **matrix of T relative to bases $\mathcal{B}$ and $\mathcal{C}$** . In general $M = [[T(b_1)]_{\mathcal{C}} \ \dots \ [T(b_n)]_{\mathcal{C}}]$.

7. Let $\mathcal{B} = \{b_1, b_2, b_3\}$ and $\mathcal{D} = \{d_1, d_2\}$ be bases for V and W respectively. Let $T : V \rightarrow W$ be the linear transformation such that $T(b_1) = 3d_1 - 5d_2$, $T(b_2) = -d_1 + 6d_2$, and $T(b_3) = 4d_2$. Find the matrix of T relative to $\mathcal{B}$ and $\mathcal{D}$.